

Two Trace-Norm Theorems for Prime-Power Lights Out Stabilization

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Abstract

For every prime $p > 5$, we prove $d(p^k - 1) = 0$ for all $k \geq 1$ whenever either $3 \mid \text{ord}_p(2)$ or $v_2(\text{ord}_p(2)) = 2$. These results use trace-norm identities in cubic norm-one groups and in the two coprime factors of $2^{2r} + 1$ for odd r . They apply to arbitrary Wieferich depth and give analytic proofs for both known Wieferich primes, 1093 and 3511, without their endpoint searches. We also derive an exact nullity-increment formula which separates new-new from new-old root collisions.

The identity $d(p^k - 1) = d(p - 1)$ for *every* prime is not proved here, and no counterexample is found. The remaining case is an arbitrary Wieferich prime with $3 \nmid \text{ord}_p(2)$ and $v_2(\text{ord}_p(2)) \neq 2$.

1 Definitions and relation to the existing work

All polynomial calculations, unless otherwise specified, take place in $\mathbb{F}_2[X]$. Let

$$F_0 = 0, \quad F_1 = 1, \quad F_{n+1} = XF_n + F_{n-1}.$$

For the bounded $n \times n$ Lights Out grid, the existing writeups use Sutner's formula

$$d(n) = \deg \gcd(F_{n+1}(X), F_{n+1}(X + 1)). \tag{1}$$

In particular $d(0) = 0$. Throughout, the exponent k in the proposed identity is positive. We write $\mu_N = \{z \in \overline{\mathbb{F}}_2 : z^N = 1\}$, $\text{ord}_m(2)$ for the multiplicative order of 2 modulo an odd integer $m > 1$, and v_ℓ for the usual ℓ -adic valuation.

The principal sources within this repository are [1, 2]. They already prove stabilization above $c_p = v_p(2^{\text{ord}_p(2)} - 1)$, give the endpoint equivalence, and analyze many unsuccessful approaches to the plateau. Section 2 supplies two additional analytic families, including both known Wieferich primes. Section 3 gives a quantitative refinement of the endpoint gcd criterion. Section 4 includes the known descent argument to make the precise remaining gap independently checkable.

For odd $N \geq 1$, define

$$H_N = F_{(N-1)/2} + F_{(N+1)/2}, \quad G_N = \gcd(H_N(X), H_N(X + 1)).$$

The Fibonacci identities give $F_N = H_N^2$, and consequently

$$d(N - 1) = 2 \deg G_N. \tag{2}$$

Here $H_1 = G_1 = 1$.

Lemma 1.1. *For odd N , the polynomial H_N is monic and square-free, with root set*

$$T_N = \{u + u^{-1} : u^N = 1, u \neq 1\} \subset \overline{\mathbb{F}}_2.$$

If $M \mid N$ is odd, then $H_M \mid H_N$.

Proof. The identity

$$(u + u^{-1})F_N(u + u^{-1}) = u^N + u^{-N}$$

shows that every displayed value is a root. For $u \neq 1$ of odd order, $u + u^{-1} \neq 0$. Two distinct nonzero elements give the same value precisely when they are mutual inverses: they are the roots of $Z^2 + (u + u^{-1})Z + 1$. Thus there are $(N - 1)/2$ distinct displayed roots, exactly the degree of the monic polynomial H_N . This proves square-freeness and the description of its roots. The inclusion $T_M \subseteq T_N$ proves divisibility. $\square$

2 Two trace–norm exclusions

By Lemma 1.1, a common root is equivalent to the existence of $u, v \neq 1$ in μ_N satisfying

$$1 + u + u^{-1} + v + v^{-1} = 0. \quad (3)$$

The exclusions below treat all pairs, including pairs of different orders and different Frobenius orbits.

2.1 The cubic norm-one group

Theorem 2.1 (Cubic exclusion). *Let q be a power of two. Suppose $u, v \in \mathbb{F}_{q^3}^\times$ have norm one to $\mathbb{F}_q$ and satisfy (3). Then either $u = 1$ and v has order 3, or $v = 1$ and u has order 3. In particular,*

$$d(q^2 + q) = 0. \quad (4)$$

Proof. Write Tr and N for the trace and norm from $\mathbb{F}_{q^3}$ to $\mathbb{F}_q$, and set $a = u + u^{-1}$. Since $N(u) = 1$,

$$a = \frac{(u + 1)^2}{u}, \quad N(a) = N(u + 1)^2.$$

Expanding the product defining $N(u + 1)$, the constant and $N(u)$ terms cancel. The sum of the pairwise products of u, u^q, u^{q^2} equals $\text{Tr}(u^{-1})$. Consequently

$$N(u + 1) = \text{Tr}(u) + \text{Tr}(u^{-1}) = \text{Tr}(a),$$

and hence

$$\boxed{N(a) = \text{Tr}(a)^2.} \quad (5)$$

Put $b = v + v^{-1} = a + 1$ and $T = \text{Tr}(a)$, and let

$$E_2 = aa^q + aa^{q^2} + a^qa^{q^2}.$$

The same identity for b , together with $\text{Tr}(1) = 1$, gives $N(b) = (T + 1)^2 = T^2 + 1$. On the other hand,

$$N(a + 1) = N(a) + E_2 + T + 1.$$

Using (5) therefore gives $E_2 = T$. The polynomial having a, a^q, a^{q^2} as its roots is thus

$$Z^3 + TZ^2 + TZ + T^2 = (Z + T)(Z^2 + T).$$

Every root of the right side belongs to $\mathbb{F}_q$, because squaring is a bijection on $\mathbb{F}_q$. In particular $a \in \mathbb{F}_q$. Now $T = a$ and $N(a) = a^3$, so (5) becomes $a^3 = a^2$. Thus $a = 0$ or $a = 1$. The equation $u + u^{-1} = 0$ forces $u = 1$, and $u + u^{-1} = 1$ forces u to have order 3. This proves the classification.

The norm-one group is μ_{q^2+q+1} . A common root for the bounded grid would require both $u, v \neq 1$, which the classification excludes. Apply Lemma 1.1 and (2). $\square$

Corollary 2.2. *If $p > 3$ is prime and $3 \mid \text{ord}_p(2)$, then*

$$d(p^k - 1) = 0 \quad \text{for every } k \geq 1,$$

whether or not p is Wieferich.

Proof. Fix k and write $f_k = \text{ord}_{p^k}(2) = f_1 p^e$, where $f_1 = \text{ord}_p(2)$ and $e \geq 0$. Set $q = 2^{f_k/3}$. Then $q^3 \equiv 1 \pmod{p^k}$. Since $p \neq 3$ and $p \nmid f_1$, the order of q modulo p is exactly 3: equivalently, $\gcd(f_1, f_k/3) = f_1/3$. Thus $q - 1$ is a unit modulo p^k , and

$$p^k \mid q^2 + q + 1.$$

Every p^k th root of unity lies in the cubic norm-one group. The group μ_{p^k} contains no element of order 3, so Theorem 2.1 excludes every relation (3). The nullity is zero. $\square$

2.2 A quadratic trace–norm law from integer factorization

Theorem 2.3 (Two-factor exclusion). *Let $r \geq 1$ be odd, and put*

$$q = 2^r, \quad s = 2^{(r+1)/2}, \quad M_{\pm} = q + 1 \pm s.$$

Suppose each of u, v belongs to $\mu_{M_-} \cup \mu_{M_+}$; the two coordinates need not belong to the same subgroup. Then (3) holds if and only if u has order 5 and $v = u^2$ or u^{-2} .

Proof. The integer identities

$$M_- M_+ = q^2 + 1, \quad \gcd(M_-, M_+) = 1 \tag{6}$$

follow from $s^2 = 2q$: the factors are odd and their difference $2s$ is a power of two. In particular, $u^{q^2} = u^{-1}$, so $a = u + u^{-1}$ belongs to $\mathbb{F}_{q^2}$. Write

$$T(a) = a + a^q, \quad N(a) = aa^q.$$

Membership in either subgroup gives $u^{q+1} = u^s$ or u^{-s} . Since s is a power of two,

$$u^{q+1} + u^{-(q+1)} = a^s.$$

Taking q th powers and using $u^{q^2} = u^{-1}$ gives

$$u^{q-1} + u^{-(q-1)} = (a^q)^s.$$

Expanding aa^q and adding these two identities proves

$$\boxed{N(a) = T(a)^s.} \tag{7}$$

This proof works for either sign in $M_{\pm}$.

The element $b = v + v^{-1}$ satisfies the same law. If $b = a + 1$, then $T(b) = T(a)$, so $N(b) = N(a)$. But

$$N(a + 1) = N(a) + T(a) + 1.$$

Hence $T(a) = 1$ and (7) gives $N(a) = 1$. It follows that

$$a^q = a + 1, \quad a(a + 1) = 1, \quad a^2 + a + 1 = 0.$$

Substitute $a = u + u^{-1}$ and multiply by u^2 to obtain

$$u^4 + u^3 + u^2 + u + 1 = 0.$$

Thus u has exact order 5. Moreover $b = a + 1 = a^2$, so the two roots of $Z^2 + bZ + 1$ are u^2, u^{-2} , giving the asserted possibilities for v . Conversely, a primitive fifth root and either of these choices satisfy (3). $\square$

Corollary 2.4. *Let p be an odd prime with $v_2(\text{ord}_p(2)) = 2$. Then, for every $k \geq 1$,*

$$d(p^k - 1) = \begin{cases} 4, & p = 5, \\ 0, & p \neq 5. \end{cases}$$

In particular, every prime $p \equiv 5 \pmod{8}$, other than 5, has zero nullity at all these levels, including any Wieferich primes in that congruence class.

Proof. The quotient $\text{ord}_{p^k}(2)/\text{ord}_p(2)$ is a power of the odd prime p . Thus $\text{ord}_{p^k}(2) = 4r$ with r odd. The unique element of order two in $(\mathbb{Z}/p^k\mathbb{Z})^\times$ is -1 , so

$$2^{2r} \equiv -1 \pmod{p^k}.$$

With the notation of Theorem 2.3, $p^k \mid M_-M_+$. Since the two factors are coprime and p^k is a prime power, it divides one factor. The theorem therefore applies to every pair in $\mu_{p^k}^2$. For $p \neq 5$ there are no solutions. For $p = 5$, only fifth roots occur, whose two trace values are the roots of $X^2 + X + 1$; the nullity is $2 \cdot 2 = 4$.

Finally, if $p \equiv 5 \pmod{8}$, then $v_2(p-1) = 2$ and Euler's criterion gives $2^{(p-1)/2} \equiv -1 \pmod{p}$. Since $\text{ord}_p(2) \mid p-1$, its two-adic valuation must be 2. $\square$

2.3 Analytic proofs for both known Wieferich primes

Corollary 2.5. *Without either endpoint search,*

$$\boxed{d(1093^k - 1) = d(3511^k - 1) = 0 \text{ for all } k \geq 1.}$$

Proof. For 1093, use $1093 \equiv 5 \pmod{8}$ and Corollary 2.4. For 3511, use $\text{ord}_{3511}(2) = 1755$, which is divisible by 3, and Corollary 2.2. A compact certificate of the latter order is

$$2^{1755} \equiv 1, \quad 2^{585} \equiv 756, \quad 2^{351} \equiv 1578, \quad 2^{135} \equiv 88 \pmod{3511}.$$

Since the prime divisors of 1755 are 3, 5, 13, these congruences prove that the order is exactly 1755. $\square$

These are not arguments that presume $c_p = 2$: both corollaries hold at every level for any value of the Wieferich depth. The factorization step in Corollary 2.4 is also where prime-power order matters. One cannot put an arbitrary divisor of M_-M_+ into one factor if it uses primes from both.

3 An exact formula for the increment

The endpoint condition is usually expressed as the triviality of a gcd. The following identity also measures the increase when that condition fails, without omitting mixed collisions.

Theorem 3.1 (Increment factorization). *Let $M \mid N$ be positive odd integers. Put*

$$K(X) = \gcd\left(\frac{H_N(X)}{H_M(X)}, H_N(X+1)\right), \quad J(X) = \gcd(K(X), K(X+1)).$$

All gcds are monic. Then

$$G_N(X) = G_M(X) \frac{K(X)K(X+1)}{J(X)} \tag{8}$$

and

$$\boxed{d(N-1) - d(M-1) = 4 \deg K - 2 \deg J.} \tag{9}$$

In particular, $d(N-1) = d(M-1)$ if and only if $K = 1$.

Proof. Write $A = T_N \setminus T_M$ and

$$C_N = T_N \cap (1 + T_N), \quad C_M = T_M \cap (1 + T_M).$$

By Lemma 1.1, these are the root sets of G_N and G_M . The root set of K is

$$S = \{\alpha \in A : \alpha + 1 \in T_N\}.$$

A point of C_N fails to belong to C_M exactly when at least one of $\alpha, \alpha + 1$ is in A . Therefore

$$C_N \setminus C_M = S \cup (1 + S).$$

The two sets on the right intersect in the root set of J . All the polynomials involved are square-free, so their union is represented by $\text{lcm}(K(X), K(X+1)) = K(X)K(X+1)/J(X)$. This union is disjoint from C_M , proving (8). Taking degrees and using (2) proves (9). Finally $\deg J \leq \deg K$, so a nonconstant K gives a strictly positive increment. $\square$

More explicitly, put

$$a = \deg J, \quad b = \deg K - \deg J.$$

There are a common roots for which both coordinates $\alpha, \alpha + 1$ are new, and b common roots with the first coordinate new and the second old. Translation supplies another b roots with the opposite orientation. Hence the increment is

$$2a + 4b.$$

For a prime-power endpoint, “new” means order above p , not necessarily one particular order when $c_p > 2$. Checking only new–new collisions cannot establish stabilization.

Remark 3.2 (A genuine mixed example, not a prime-power counterexample). *Direct polynomial arithmetic for $M = 17$ and $N = 255$ gives*

$$d(16) = 8, \quad d(254) = 140, \quad \deg K = 62, \quad \deg J = 58.$$

Thus $a = 58, b = 4$, and the increment is $2 \cdot 58 + 4 \cdot 4 = 132$. An internal-collision calculation alone would miss the contribution 16 from the mixed pairs. Since $255 = 3 \cdot 5 \cdot 17$, this example does not contradict the prime-power conjecture.

One explicit mixed pair can be checked in $\mathbb{F}_2[t]/(t^8 + t^4 + t^3 + t^2 + 1)$, where t has order 255:

$$\begin{aligned} u &= t^{17}, \quad v = t^{120}, \quad \text{ord}(u) = 15, \quad \text{ord}(v) = 17, \\ u + u^{-1} &= t^7 + t^4 + t + 1, \quad v + v^{-1} = t^7 + t^4 + t. \end{aligned}$$

Thus the two traces differ by one, even though only the second coordinate has order dividing 17.

4 Exactly what is already proved

For an odd prime p , write

$$f = \text{ord}_p(2), \quad c = v_p(2^f - 1), \quad \rho_i = \min\{r \geq 1 : 2^r \equiv 1 \text{ or } -1 \pmod{p^i}\}.$$

The order-lifting formula and cyclicity of $(\mathbb{Z}/p^i\mathbb{Z})^\times$ give

$$\text{ord}_{p^i}(2) = fp^{\max(0, i-c)}, \quad \rho_i = \rho_1 p^{\max(0, i-c)}. \quad (10)$$

Indeed, a multiple ft of f satisfies $v_p(2^{ft} - 1) = c + v_p(t)$. The signed order is the ordinary order divided by two when the ordinary order is even, and is the ordinary order otherwise.

If u has exact order p^i , then

$$[\mathbb{F}_2(u + u^{-1}) : \mathbb{F}_2] = \rho_i. \quad (11)$$

To see this, Frobenius to the power r fixes $u + u^{-1}$ exactly when u^{2^r} is u or u^{-1} .

Theorem 4.1 (Known descent above the plateau). *Every common root of $F_{p^k}(X)$ and $F_{p^k}(X + 1)$ already belongs to $T_{p^{\min(k,c)}}$. In particular,*

$$d(p^k - 1) = d(p^c - 1) \quad (k \geq c).$$

Proof. Write a common root and its translate as

$$\alpha = u + u^{-1}, \quad \alpha + 1 = v + v^{-1},$$

where $u, v \neq 1$ have exact orders p^i, p^ℓ , respectively. Since $\mathbb{F}_2(\alpha) = \mathbb{F}_2(\alpha + 1)$, (10)–(11) imply either $i, \ell \leq c$, or $i = \ell > c$.

Suppose $i = \ell > c$. Set $L = p^i$ and $m = p^{i-c}$. Taking u as a primitive L th root, write $v = u^s$ with $p \nmid s$ and $1 \leq s < L$. Then u is a zero of

$$P(X) = 1 + X + X^{L-1} + X^s + X^{L-s}.$$

Let h be the minimal polynomial of u^m over $\mathbb{F}_2$. Its degree is f , while the degree of u is fm . Therefore $h(X^m)$ is the minimal polynomial of u , and divides P .

Decompose polynomials uniquely by their exponent classes modulo m :

$$P(X) = \sum_{r=0}^{m-1} X^r P_r(X^m).$$

Divisibility by $h(X^m)$ implies $h \mid P_r$ for every r . But none of $1, -1, s, -s$ is zero modulo m , so $P_0 = 1$. This is impossible for the nonconstant polynomial h . Thus both orders are at most p^c . The root sets are nested and every root of F_{p^k} has multiplicity two, proving the stated equality of nullities. $\square$

Corollary 4.2. *The desired identity holds for every non-Wieferich odd prime. It also holds for $p = 2$.*

Proof. For a non-Wieferich odd prime, $c = 1$, so apply Theorem 4.1. For $p = 2$, $F_{2^k}(X) = X^{2^k-1}$ is coprime to its translate. Thus $d(2^k - 1) = d(1) = 0$. $\square$

5 The remaining assertion, with both types of collision

Unresolved assertion 5.1 (Wieferich endpoint). *For every Wieferich prime p , with $c = v_p(2^{\text{ord}_p(2)} - 1)$, prove*

$$K_p(X) := \gcd\left(\frac{H_{p^c}(X)}{H_p(X)}, H_{p^c}(X + 1)\right) = 1. \quad (12)$$

By Theorems 3.1 and 4.1, this assertion is equivalent to the original all-prime identity. It is not a consequence of those theorems. In fact, they give the exact possible discrepancy

$$d(p^c - 1) - d(p - 1) = 4 \deg K_p - 2 \deg \gcd(K_p(X), K_p(X + 1)).$$

By Corollaries 2.2 and 2.4, the unresolved primes can now be restricted further to

$$c_p \geq 2, \quad 3 \nmid \text{ord}_p(2), \quad v_2(\text{ord}_p(2)) \neq 2. \quad (13)$$

No claim that this set is empty is made.

For $p > 3$, an equivalent finite-field formulation is

$$u, v \in \mu_{p^c}, \quad 1 + u + u^{-1} + v + v^{-1} = 0 \implies u, v \in \mu_p. \quad (14)$$

Neither coordinate can equal 1: that would force the other to have order 3. Thus (14) is precisely the common-root condition of Lemma 1.1. Notice that the hypothesis allows different orders for u and v .

5.1 Why the established proof does not extend through the plateau

For $1 \leq i \leq c$, all groups μ_{p^i} lie in the same field $\mathbb{F}_{2^f}$, and their primitive traces all have degree ρ_1 . Equal degrees no longer imply equal levels. In the residue-class proof, the proposed substitution has $m = 1$, so its decisive class P_0 is the whole polynomial, not the constant 1.

There is also no field-theoretic descent obtained simply by taking p th powers. On μ_{p^i} this operation has a kernel of order p ; it is not a field automorphism. The fact that u^p, v^p have smaller orders does not imply that they satisfy the same additive equation. Likewise, a norm identity does not make all conjugates of an element vanish at a chosen prime above two. On the plateau the cyclotomic extension splits at two, rather than supplying the nontrivial residue field extension used by the off-plateau proof.

5.2 What a counterexample certificate must contain

A counterexample can be presented without constructing a large grid. Choose a prime $p > 3$, an integer $2 \leq j \leq c_p$, an irreducible $h \in \mathbb{F}_2[T]$, and an integer b with $1 \leq b < p^j$. In $\mathbb{F}_2[T]/(h)$, let t be the residue of T and verify

$$t^{p^j} = 1, \quad t^{p^{j-1}} \neq 1, \quad 1 + t + t^{p^j-1} + t^b + t^{p^j-b} = 0.$$

These equalities exhibit a new common root and imply $d(p^j - 1) > d(p - 1)$. Conversely, from any counterexample choose a coordinate of largest order as t ; this produces such a certificate. One must not require $p \nmid b$, since a mixed-order counterexample could have $p \mid b$.

6 Exact computations and their scope

The existing search in [3] uses characteristic-two field arithmetic, not the conjectured prime-power reduction. Its 64-bit fingerprint is an additive projection: an exact equality necessarily survives projection. Thus an empty fingerprint-candidate set is a rigorous absence certificate, not a probabilistic conclusion. Any surviving candidates are checked in the full field.

Both known Wieferich endpoint searches were rerun in this investigation:

p	c_p	$\text{ord}_p(2)$	fingerprint candidates	$d(p^2 - 1) = d(p - 1)$
1093	2	364	0	0
3511	2	1755	0	0

The depth $c_p = 2$ is verified by $2^{\text{ord}_p(2)} \equiv 1 \pmod{p^2}$ but not modulo p^3 . The zero endpoint nullities imply zero base nullities by root-set inclusion. Reproducible commands from the repository root are

```
python code\ai_prototypes\five_term_char2.py 1093 2
python code\ai_prototypes\five_term_char2.py 3511 2
```

These are now independent checks of Corollary 2.5, not prerequisites for it.

The trace–norm laws were also checked directly in finite fields. For the cubic law, $q = 2, 4, 8, 16, 32$ gives only the classified degenerate relations: respectively $0, 4, 0, 4, 0$ ordered pairs. For the two-factor law, $r = 1, 3, 5, 7$ gives $(M_-, M_+) = (1, 5), (5, 13), (25, 41), (113, 145)$. In each union of the two subgroups there are exactly eight ordered relations, all from the fifth roots as proved above.

For small examples, direct Fibonacci gcd calculations give

p	$d(p-1)$	$d(p^2-1)$	$d(p^3-1)$
3	0	0	0
5	4	4	4
7	0	0	0
11	0	0	0
17	8	8	8
31	20	20	20

These calculations use `_adjacent_fibonacci_pair` and `GF2Polynomial.gcd`, not the optimized `grid_nullity` shortcut for proven prime-power cases. The factorization in Theorem 3.1 was also checked directly for every proper odd divisor pair $M \mid N$, $3 \leq N \leq 301$: 398 pairs, including 13 with a nonzero mixed contribution. The proof of the theorem is the root-set argument, not this finite check.

7 Conclusion

Two arithmetic classes are now covered without a non-Wieferich hypothesis: $3 \mid \text{ord}_p(2)$ and $v_2(\text{ord}_p(2)) = 2$. In particular, both known Wieferich primes have analytic all-level proofs, rather than only computational endpoint proofs. The increment factorization additionally measures both kinds of possible new collision.

The all-prime question nevertheless remains unresolved by this investigation. The missing step is (12), equivalently (14), for a Wieferich prime satisfying (13). Neither trace-norm theorem addresses all of those primes, and no counterexample certificate of the required kind has been found.

Further attempts at conductor descent, rather than additional zero-nullity families, are recorded in `tex\prime_power_endpoint\conductor_descent.tex`. That addendum gives an obstruction to direct powering in the regular-map formulation and a single-candidate-phase criterion for mixed collisions when H_p is irreducible; it does not claim a general proof.

References

- [1] W. Boyles, *Fibonacci Polynomials & Lights Out*. Repository manuscript: `tex\finite_fields\finite_fields.tex`. See “Orders of Prime Powers” and “Applications to $d(n)$ ”.
- [2] W. Boyles, *Reciprocal Five-Term Relations among p^j -th Roots of Unity in Characteristic Two: Reductions, Obstructions, and the Wieferich Plateau*. Repository manuscript: `tex\five_term_plateau\five_term_plateau.tex`. See “The single endpoint statement”, “Two collision norms”, and “Computations”.
- [3] Repository implementations: `code\kernel_size.py`, `code\polynomials.py`, `code\ai_prototypes\five_term_char2.py`, and `code\ai_prototypes\wieferich_check.py`.